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Project: Power Bill: Residential Rates

Company: Alabama Power

Website: <https://www.alabamapower.com/content/dam/alabama-power/pdfs-docs/Rates/FD.pdf>

Objectives:

- (1.) Calculate the June through September power bill of the residents of Alabama within each range of specific water usage manually using Arithmetic method.
- (2.) Write a piecewise function of the resident rates.
- (3.) Recalculate the same power bill of resident of Alabama within each range of specific usage algebraically using the piecewise function method.

Information:

Monthly Rate

Base Charge: \$14.50 per customer; plus

Charge for Energy:

Billing Months June-September

12.4384¢ per kWh for the first 1000 kWh, plus

12.6913¢ per kWh for all over 1000 kWh

Conversions (From cents to dollars):

$$12.4384\text{¢} = \frac{\$12.4384}{100} = \$0.124384$$

$$12.6913\text{¢} = \frac{\$12.6913}{100} = \$0.126913$$

Arithmetic Method:

(1.) 0 kWh

(2.) 700 kWh

(3.) 1500 kWh

(1.) 0 kWh

$$\text{cost} = \$14.50$$

(2.) 700 kWh

$$\text{cost} = 14.50 + 0.124384(700)$$

$$\text{cost} = \$101.5688$$

$$\text{cost} \cong \$101.57$$

(3.) 1500 kWh

$$\text{cost} = 14.50 + 0.124384(1000) + 0.126913(1500 - 1000)$$

$$\text{cost} = 14.50 + 0.124384(1000) + 0.126913(500)$$

$$\text{cost} = \$202.3405$$

$$\text{cost} \cong \$202.34$$

Piecewise Function

Let us define variables

$p = \text{power consumed in kWh}$

$c(p) = \text{cost of power consumed in \$}$

first piece : $0 \leq p \leq 1000$

$$c(p) = 14.50 + 0.124384p$$

$$c(p) = 0.124384p + 14.50$$

first piece : $0 \leq p \leq 1000$

second piece : $p > 1000$

$$c(p) = 14.50 + 0.124384(1000) + 0.126913(p - 1000)$$

$$c(p) = 14.50 + 124.384 + 0.126913p - 126.913$$

$$c(p) = 0.126913p + 11.971$$

$$c(p) = \begin{cases} 0.124384p + 14.50; & 0 \leq p \leq 1000 \\ 0.126913p + 11.971; & p > 1000 \end{cases}$$

Piecewise Function Method

Let us test this Piecewise Function with the three values we used for the Arithmetic Method.

(1.) 0 kWh

This falls in the first piece.

$$c(p) = 0.124384p + 14.50$$

$$c(0) = 0.124384(0) + 14.50$$

$$c(0) = 0 + 14.50$$

$$c(0) = \$14.50$$

(2.) 700 kWh

This falls in the first piece.

$$c(p) = 0.124384p + 14.50$$

$$c(700) = 0.124384(700) + 14.50$$

$$c(700) = 87.0688 + 14.50$$

$$c(700) = \$101.5688$$

$$c(700) \cong \$101.57$$

(3.) 1500 kWh

This falls in the second piece.

$$c(p) = 0.126913p + 11.971$$

$$c(1500) = 0.126913(1500) + 11.971$$

$$c(1500) = 190.3695 + 11.971$$

$$c(1500) = \$202.3405$$

$$c(1500) \cong \$202.34$$

References (MLA citation):

Chukwuemeka, Samuel. "Where Is the Light Bill? It Is a Piecewise Function!"

Where Is the Light Bill? It Is a Piecewise Function!,

conferencepresentations.appspot.com/Projects/PreCalculus/PiecewiseFunctions/PowerBill.html. Accessed 24 Oct. 2024.

Rates FD Family Dwelling - Alabama Power,

www.alabamapower.com/content/dam/alabama-power/pdfs-docs/Rates/FD.pdf. Accessed 24 Oct. 2024.